

SITTTR KERALA — REVISION 2021

CAD Lab

Course Code: 2078

Semester: 2

Chemical Engineering

[\[Official Syllabus Reference\]](#)

Course Overview

Program	Diploma in Chemical Engineering	Credits	0
Course Category	Engineering Science	Total Hours	45 Periods
Teaching Scheme	L:0 T:0 P:3	Assessment	CIE & ESE (Internal/External Lab)

The CAD Lab course is designed to provide hands-on experience in Computer Aided Design and Drafting using industry-approved software (AutoCAD). It bridges the gap between manual engineering graphics and modern digital drafting standards.

Course Outcomes (COs)

CO No.	Description	Cognitive Level
CO1	Summarize Computer Aided Design.	Understanding
CO2	Illustrate the basic commands in AutoCAD.	Applying
CO3	Sketch block and Isometric 2D drawing in AutoCAD.	Applying
CO4	Illustrate 3D modelling in AutoCAD.	Applying

Module 1: Introduction to CAD

Foundations and Industry Context (3 Hours)

CAD (Computer-Aided Design) is the use of computer systems to assist in the creation, modification, analysis, or optimization of a design.

Advantages of CAD:

- **Accuracy:** Extremely high precision compared to manual drafting.
- **Speed:** Faster creation and editing of drawings.
- **Storage:** Drawings can be stored digitally and easily shared.
- **Standardization:** Easy to maintain uniform line weights, fonts, and styles.
- **Reusability:** Components (Blocks) can be reused across multiple projects.

Malayalam Support: CAD ഓഫീസിൽ പരമ്പരാഗതമായി ഉപയോഗിക്കുന്ന സോഫ്റ്റ്‌വെയർ ആണ്. ഇത് ഓട്ടോമേറ്റഡ് ഡ്രോയിംഗ് സോഫ്റ്റ്‌വെയർ ആണ്. ഇത് ഓട്ടോമേറ്റഡ് ഡ്രോയിംഗ് സോഫ്റ്റ്‌വെയർ ആണ്.

Commonly used CAD software in industry: AutoCAD, SolidWorks, CATIA, MicroStation, Revit.

AutoCAD Drawing Editor Components:

- **Ribbon:** Contains toolsets organized by tabs.
- **Command Line:** The primary interface for entering commands and viewing prompts.
- **Status Bar:** Controls for drawing aids like Ortho, Snap, and Grid.
- **Drawing Area:** The canvas where designs are created.
- **UCS Icon:** Shows the orientation of X and Y axes.

Module 2: Basic Commands & 2D Drafting

The Core of AutoCAD Operations (21 Hours)

Before drawing, the environment must be set up:

- **UNITS:** Sets the measurement system (Architectural, Decimal, Engineering).
- **LIMITS:** Defines the boundaries of the drawing area.
- **ZOOM:** Controls the display magnification (Zoom All, Zoom Window).
- **SNAP & GRID:** Aids for precise positioning.
- **Function Keys:** F3 (OSNAP), F7 (GRID), F8 (ORTHO).

Setup Procedure:

1. Type UNITS → Set to Decimal, Precision 0.00.
2. Type LIMITS → Lower Left (0,0), Upper Right (420,297 for A3).
3. Type z → A (Zoom All).

Drawing Commands:

- **LINE (L):** Creates straight line segments.
- **CIRCLE (C):** Creates circles using Center/Radius or Diameter.
- **POLYLINE (PL):** Creates a single object composed of line and arc segments.
- **HATCH (H):** Fills enclosed areas with patterns (useful for sectional views).

Modify Commands:

- **OFFSET (O):** Creates concentric circles or parallel lines.
- **TRIM (TR):** Cuts objects at a cutting edge.
- **MIRROR (MI):** Creates a reflected copy of objects.
- **FILLET (F):** Rounds the edges of objects.

Malayalam Support: Modify commands to support Malayalam text processing. For example, use the `trim` command to remove leading and trailing spaces from Malayalam text.

Layers: Used to organize drawings by grouping objects (e.g., Hidden lines, Dimensions, Text). You can control color, linetype, and visibility (Freeze/Thaw).

Dimensioning: Essential for engineering drawings. Commands include DIMLINEAR, DIMANGULAR, and DIMRADIUS.

Layers: Used to organize drawings by grouping objects (e.g., Hidden lines, Dimensions, Text). You can control color, linetype, and visibility (Freeze/Thaw).

Dimensioning: Essential for engineering drawings. Commands include DIMLINEAR, DIMANGULAR, and DIMRADIUS.

Module 3: Blocks & Isometric Drawing

Efficiency and 2D Projections (9 Hours)

Module 3: Blocks & Isometric Drawing

Efficiency and 2D Projections (9 Hours)

A **Block** is a collection of objects that are combined into a single named object. Blocks are used for repetitive symbols (valves, pumps in Chemical Engineering).

- **BLOCK (B):** Creates a block definition.
- **INSERT (I):** Places a block into the drawing.

- A **Block** is a collection of objects that are combined into a single named object. Blocks are used for repetitive symbols (valves, pumps in Chemical Engineering).
- **BLOCK (B):** Creates a block definition.
 - **INSERT (I):** Places a block into the drawing.

- **WBLOCK:** Saves a block as a separate file for use in other drawings.

Isometric drawing is a 2D representation of a 3D object where the axes are at 120° to each other.

- **ISOPLANE:** Switches between Left, Top, and Right planes (F5 key).
- **ISOCIRCLE:** Used within the ELLIPSE command to draw circles in isometric view.

Malayalam Support: Isometric drawing അളവുകൾ 3D വസ്തുവിന്റെ 2D പ്രതിരോധം അളവുകൾ അളവുകൾ. F5 കീ അളവുകൾ അളവുകൾ അളവുകൾ.

Module 4: 3D Modelling

Solid Modeling Fundamentals (12 Hours)

To start 3D, switch the workspace to "3D Modeling". Use the **ViewCube** or **3DORBIT** to navigate.

Viewport: Divides the screen into multiple views (Top, Front, Side, Isometric) to work on 3D objects simultaneously.

- **EXTRUDE:** Creates a 3D solid from a 2D face by adding height.
- **PRESSPULL:** Dynamically adds or subtracts volume by selecting an enclosed area.
- **REVOLVE:** Creates a 3D solid by rotating a 2D profile around an axis (ideal for pipes/tanks).
- **3D FILLET:** Rounds the edges of a 3D solid.

Creating a Cylinder:

1. Draw a Circle in 2D.
2. Type EXTRUDE.
3. Select the circle and specify the height.

Coordinate Entry Guide

AutoCAD uses three main methods for entering coordinates:

Absolute: (X, Y)

Example: 10,20

Relative: @X, Y

Example: @50,0

Polar: @Distance < Angle

Example: @100<45

Diagram Library

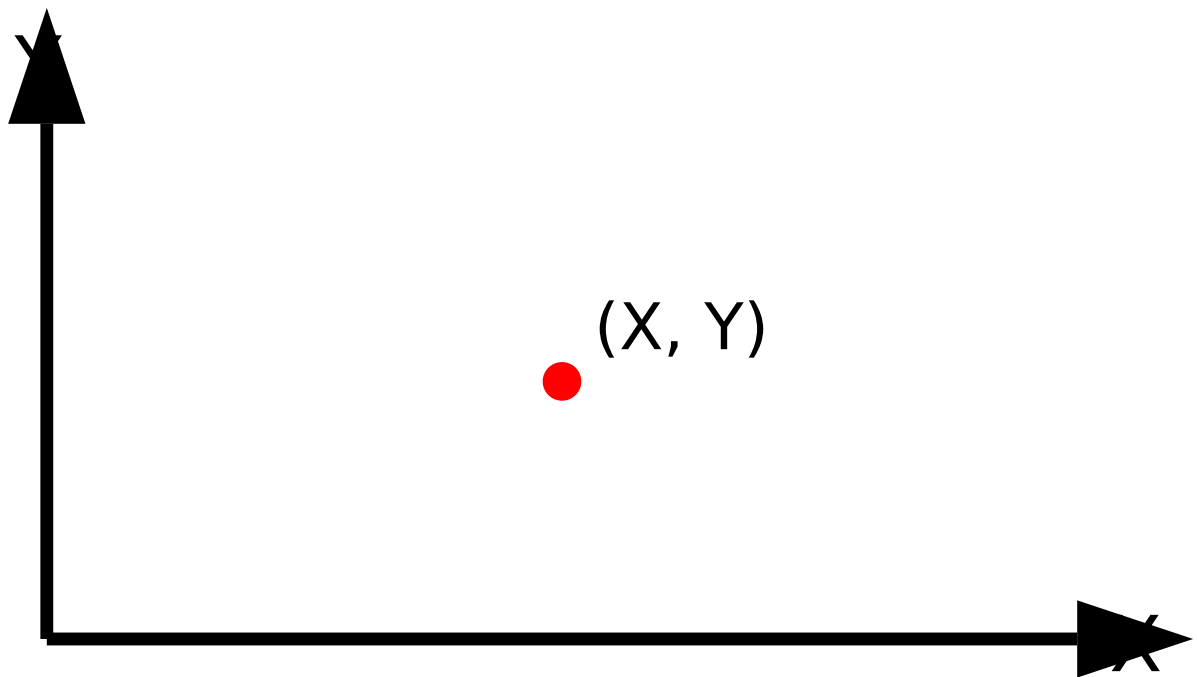


Figure 1: Cartesian Coordinate System in AutoCAD

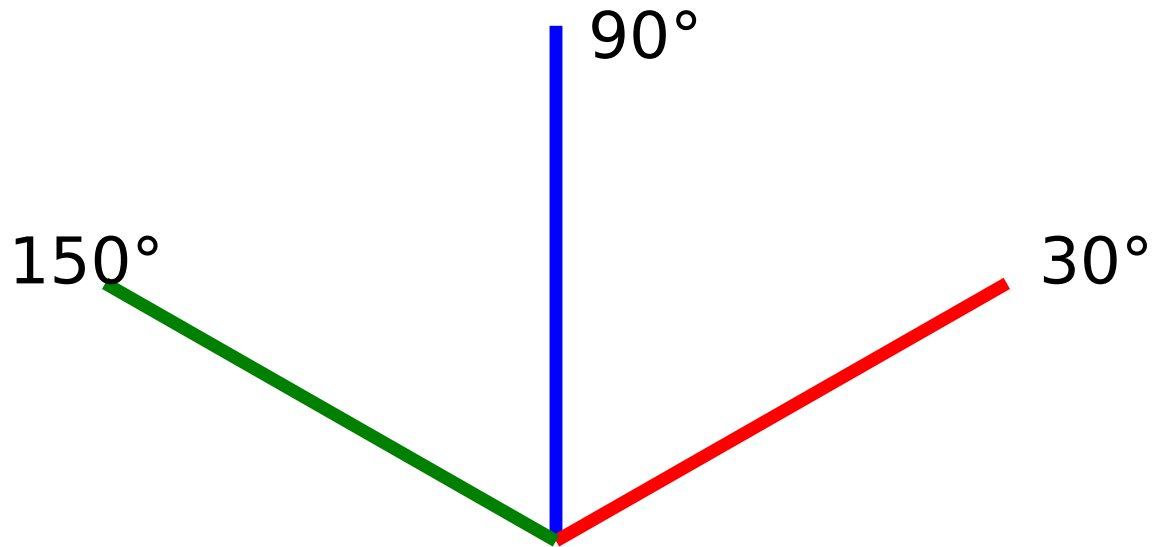


Figure 2: Isometric Axis Orientation

Practice Questions & Answers

What is the difference between 'Freeze' and 'Off' in Layers?

Off: Makes the layer invisible but AutoCAD still calculates the geometry during regenerations.

Freeze: Makes the layer invisible and AutoCAD ignores it during regenerations, improving performance in large drawings.

How do you draw a circle in an Isometric plane?

1. Ensure ISODRAFT is ON.
2. Type ELLIPSE (EL).

3. Choose the **Isocircle** option.
4. Specify center and radius.

Explain the 'Presspull' command.

Presspull is used to create or modify 3D solids. By clicking inside a bounded 2D area, you can pull it to add volume or push it to create a hole/subtract volume.

Practice Model Paper

Time: 3 Hours | Max Marks: 100

1. **Part A (Basic):** Set up an A4 sheet with decimal units and 0.0 precision. Draw a title block at the bottom right. (20 Marks)
2. **Part B (2D):** Draw the sectional front view of a Flanged Pipe Joint using appropriate layers and hatching. (40 Marks)
3. **Part C (3D):** Create a 3D model of a simple Storage Tank with a conical roof using REVOLVE or EXTRUDE commands. (40 Marks)

Quick Revision: Keyboard Shortcuts

L	Line	CO	Copy
C	Circle	M	Move
TR	Trim	RO	Rotate

O	Offset	SC	Scale
F	Fillet	EX	Extend
H	Hatch	B	Block

- **GUI:** Graphical User Interface.
- **Ortho Mode:** Restricts cursor movement to horizontal and vertical directions.
- **OSNAP:** Object Snap (snaps to endpoints, midpoints, etc.).
- **UCS:** User Coordinate System.
- **WCS:** World Coordinate System (Fixed).

Source URL: [SITTTR Syllabus Portal](#)

1. **2D** **புள்ளிப்போட்டி**: **புள்ளி**, **புள்ளிப்போட்டி** **புள்ளிப்போட்டி** **புள்ளிப்போட்டி** **புள்ளிப்போட்டி** **புள்ளிப்போட்டி**.
2. **புள்ளிப்போட்டி**: **புள்ளி** **புள்ளிப்போட்டி** **3D** **புள்ளிப்போட்டி** **2D-புள்ளி** **புள்ளிப்போட்டி**.
3. **3D** **புள்ளிப்போட்டி**: **புள்ளிப்போட்டி** (**Extrude**), **புள்ளிப்போட்டி** (**Revolve**) **புள்ளிப்போட்டி** **புள்ளிப்போட்டி**

□□□□□□ 3D □□□□□□ □□□□□□□□□□.

□□□ □□□□□□ □□□□□ □□□□□ □□□□ 'Enter' □□□□□□□□□□□□.